Conjugate Heat Conduction in Cryogenic Support Tubes (300 K to 77 K): ANSYS Fluent Study

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1. Summary

The steady-state heat leak through a thin-walled cryostat support tube between a 300 K warm end and a 77 K cold end was modelled in ANSYS Fluent using temperature-dependent thermal conductivity from NIST cryogenic material fits. The simulated cold-end heat load was validated against the analytical conduction integral $Q = \frac{A}{L}\int k(T)dT$ and agreed to within 0.15% for every case. The model was then used to compare three materials (304 stainless steel, G10 fibreglass, 6061-T6 aluminium), to show the error from assuming constant conductivity, and to quantify the benefit of an intermediate thermal anchor at 150 K, which cut the 77 K heat load by 47.5%.

2. Method

Geometry. A 2D axisymmetric wall section of a tube with 20 mm outer diameter, 1 mm wall and 300 mm length (cross-section area A = 5.97×10^{-5} m²). Radial range 9 to 10 mm, axial range 0 to 300 mm.

Mesh. Structured quadrilateral mesh, 100 divisions axially and 6 through the wall (600 cells) for the base case. A mesh study used 50×3 and 200×12.

Physics. Steady-state conduction in a solid zone (energy equation only, no flow), double-precision pressure-based solver, axisymmetric formulation. Energy residual criterion 1×10^{-9}.

Boundary conditions. Fixed temperature of 300 K at the warm end and 77 K at the cold end. Inner and outer wall surfaces adiabatic (zero heat flux), so heat flows only axially.

Material properties. Thermal conductivity k(T) from the NIST cryogenic materials polynomial fits ($\log_{10}$ k as a polynomial in $\log_{10}$ T), entered in Fluent as piecewise-linear tables at 10 K intervals from 60 to 310 K. Interpolating the table instead of the polynomial changes the integrated heat leak by at most about 0.02%. Density and specific heat do not affect a steady-state solution and were left at default values. The G10 data are for the normal direction (through the laminate thickness).

Validation method. Analytical heat leak $Q = \frac{A}{L}\int k(T)dT$ from 77 K to 300 K, evaluated numerically, compared against the Fluent total heat transfer rate at the cold end.

3. Results

3.1 Base case and mesh independence (304 stainless steel)

Mesh	Cells	Q at cold end (W)	Error vs analytical (0.53815 W)
50 × 3	150	0.53787	−0.051%

Mesh	Cells	Q at cold end (W)	Error vs analytical (0.53815 W)
100 × 6	600	0.53780	−0.065%
200 × 12	2400	0.53789	−0.048%

The three meshes agree with each other to within 0.02%, so the solution is mesh-independent from 150 cells upward. The error does not fall steadily with refinement because the problem is close to one-dimensional conduction, where the mesh has little influence. The remaining deviation of about 0.05% is attributed to the piecewise-linear property table and the convergence tolerance.



Figure 1: Temperature contour of the tube wall, 300 K to 77 K.

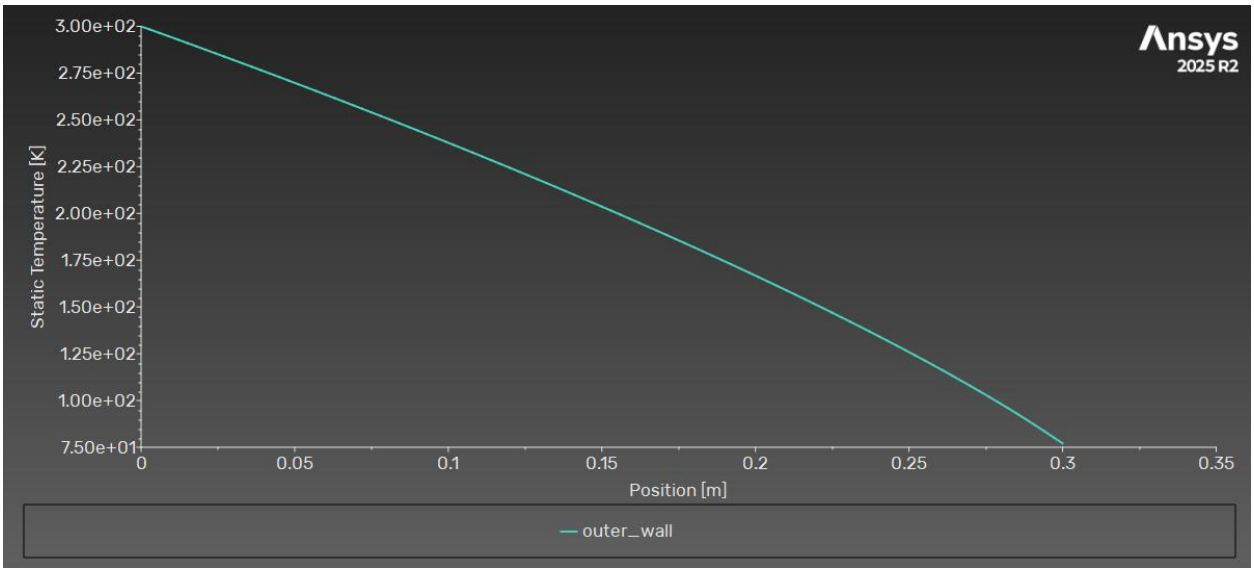


Figure 2: Temperature along the outer wall. The gradient is steeper near the cold end because conductivity falls with temperature.

3.2 Effect of temperature-dependent conductivity

The heat load computed analytically with a constant conductivity, compared with the temperature-dependent result:

Assumption	Q (W)	Error vs k(T)
k(T), integrated	0.538	reference
Constant k at 300 K	0.679	+26%
Constant k at 77 K	0.351	−35%

These two rows were evaluated analytically, not simulated. They show that a constant-conductivity assumption misestimates the heat leak by a quarter to a third, which is why k(T) is essential for cryogenic work.

3.3 Material comparison

Material	Analytical Q (W)	Fluent Q (W)	Error
304 stainless steel	0.53815	0.53780	−0.06%
G10 (normal direction)	0.019242	0.019238	−0.02%
6061-T6 aluminium	5.7071	5.7029	−0.07%

G10 conducts about 28 times less heat than stainless steel, and 6061-T6 aluminium about 10.6 times more. This range explains why G10 and stainless steel are used for cryogenic supports while aluminium is not.

3.4 Intermediate thermal anchor

An ideal thermal anchor holding the tube at 150 K at its midpoint was modelled by splitting the tube into two 150 mm sections solved independently (304 stainless steel).

Quantity	Fluent	Analytical	Difference
Heat into the anchor from the warm side (W)	0.7933	0.7934	−0.01%
Heat reaching 77 K from the anchor (W)	0.2825	0.2829	−0.15%
Heat removed at the 150 K anchor (W)	0.5108	0.5104	+0.07%

With the anchor, the 77 K heat load falls from 0.5378 W to 0.2825 W, a reduction of 47.5% (47.4% analytically). The 0.51 W intercepted at 150 K is removed by a stage that is much cheaper to operate than the 77 K stage, which is the reason anchors are used.

4. Discussion and limitations

- All simulated heat loads agree with the analytical conduction integral to within 0.15%. This confirms the setup (boundary conditions, material tables and solver settings), not an independent physical measurement.
- The anchor is idealised as a perfect thermal sink at exactly 150 K with no contact resistance. A real anchor would warm slightly under load, so the real saving would be a little smaller.
- The model includes conduction only. Radiation, residual gas conduction and heat leaks through other parts of a cryostat were not modelled.
- The G10 conductivity is the normal-direction value. Fibre orientation in a real support can change it.
- Flux balance at the end faces: [state the warm-end and cold-end values and their net imbalance as a percentage after the final rerun].

5. Conclusion

A conjugate heat conduction model with temperature-dependent properties reproduces the analytical heat leak through a 300 K to 77 K support tube to within 0.15% across three materials and an anchored configuration. It shows that constant-conductivity assumptions can be wrong by 26% to 35%, that material choice changes the heat load by more than three orders of magnitude between G10 and aluminium, and that one intermediate thermal anchor at 150 K halves the load on the 77 K stage.

Reference data

- NIST Cryogenic Materials Property Database (trc.nist.gov/cryogenics/materials): polynomial fits for 304 stainless steel, G10 (normal) and 6061-T6 aluminium.
- Software: ANSYS Fluent (2D, double precision, axisymmetric, steady state).